

CS4.406 Assignment 1 — Lexical & Semantic Retrieval: Design Note

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1 What I built

Raw archives → per-dataset readers → **one unified schema** → temporal split → parquet feature store (42s rebuild, both datasets). Two retrievers behind one **Retriever** protocol — BM25 over title+abstract, and MiniLM embeddings with a brute-force exact index — plus recency, popularity, random, RRF fusion and a popularity-prior blend. One `evaluate()` scores all of them, so Q4.5 holds by construction: there is nowhere else a metric can be computed.

Two evaluation regimes are kept strictly apart, because conflating them produces numbers that look fine and mean nothing. *Corpus regime*: recall@K over all 21K–65K articles — *did the clicked article survive the cut?* *Slate regime*: AUC/MRR/nDCG over the ~11–40 candidates actually shown — *did you rank it above the others?* Every result row carries a **regime** column.

2 The constraint that drove every design choice

Every richer feature EB-NeRD offers is one MIND lacks: body text, click timestamps, publish time, session ids (0% on MIND), provided embeddings. Using any of them for a headline number silently converts a *dataset* comparison into a *feature-availability* comparison.

Rule adopted: the shared path uses only the intersection — title, abstract, category, click history. Everything one-sided becomes a labelled single-dataset ablation. This one rule resolves the encoder choice, recency weighting, body text and session context together.

3 Key decisions and rejected alternatives

Decision	Chosen	Rejected, and why
Corpus	Union of MIND’s two <code>news.tsv</code> (65,238)	Per-split: 23.3% of dev clicks are on train-absent articles, an invisible 0.767 recall ceiling [F2,F20]
Split	Hold out official test, carve val from train tail	Literal “7 days test”: MIND’s <i>entire</i> labelled span is 6.0 days [F27]
Lexical fields	Title + abstract (Q2.1 mandates)	Title+body — MIND has no body, breaking comparability [F10]
Encoder	MiniLM 384-d , mean-pooled, L2-normalised	XLNet-R fails the Danish probe (below); [CLS] pooling; per-language stemming
Index	Brute force (exact; Q3.2 permits it)	HNSW — both corpora are <200K, where exact costs 190–890 ms and is the recall <i>ceiling</i> ANN is measured against
User vector	Log positional decay, $n=20$, conditional pooling	Exponential decay collapses the query to 2–3 clicks; plain mean blurs multi-interest users
Session context	Dropped , deferred to C-2	0% coverage on MIND; EB-NeRD sessions average 1.9 impressions [F29]
Coverage CI	Point estimate, marked (no CI)	Every bootstrap scheme is biased low — see §4.6

4 Observations

4.1 Recency dominates; text matching is secondary [F16,F21]

On EB-NeRD, **94.3% of clicks are on articles under 24 h old** while only ~1.1% of the corpus is that fresh. A retriever that *ignores the user entirely* and returns the newest articles scores **recall@50 = 0.8986**, against BM25’s 0.0043 over the full corpus. Restricting either retriever to a 24 h window moves it from ~0.004 to ~0.24 — a **33× swing from the candidate pool, not the scoring function**.

Candidate generation on news is primarily a *freshness* problem. This is legitimate, not a leak: publish time is known at serving time; ranking by *future* engagement would not be.

4.2 Lexical vs. semantic: the answer flips between datasets [Q3.5]

recall@100, $n=4,000$	EB-NeRD (Danish)	MIND (English)
Best overall	recency 0.9434 [0.9361, 0.9505]	popularity 0.0688 [0.0615, 0.0763]
Lexical	bm25+24h 0.2375 [0.2244, 0.2490]	bm25 0.0142 [0.0108, 0.0172]
Semantic	sem+24h 0.2307 [0.2175, 0.2432]	sem 0.0163 [0.0130, 0.0199]
RRF fusion	0.0070 — <i>no gain</i>	0.0181 [0.0146, 0.0218] — <i>best retriever</i>

On MIND semantic beats lexical and fusion beats both; on EB-NeRD lexical edges ahead and fusion helps nothing. Fusion exploits disagreement: on EB-NeRD the two retrievers agree (CIs overlap heavily), so there is nothing to fuse. A single “which is better” answer would be wrong.

The slice tells a sharper story. EB-NeRD recall@100, cold defined as history ≤ 97 (q0.25 of observed; median 235):

	cold ($n=1,004$)	warm ($n=2,996$)
semantic + 24h	0.2468 [0.2198, 0.2742]	0.2253 [0.2104, 0.2407]
bm25 + 24h	0.2216 [0.1982, 0.2490]	0.2428 [0.2276, 0.2588]

The ordering crosses over: semantic wins cold users, lexical wins warm ones. With few clicks, word overlap has too few terms to work with while embeddings still place a sparse history somewhere meaningful. CIs overlap, so this is suggestive, not established — but the crossover is the shape, and it is what Q3.5 asks for.

4.3 XLM-RoBERTa cannot separate related from unrelated Danish [F37]

The brief names BERT/XLM-RoBERTa. Before trusting either, a probe embedded 5 obviously-related Danish headline pairs and 5 obviously-unrelated ones:

Encoder	related	unrelated	margin	verdict
xlm-roberta-base (768-d)	0.9972	0.9954	+0.0018	OVERLAPS
MiniLM (384-d)	0.6523	0.0253	+0.6271	SEPARATES

XLM-R rates “Brøndby beat FCK” and “a new apple cake recipe” as 0.995 similar. It is trained for masked-language modelling, not to make cosine similarity meaningful, and its vectors collapse into a narrow cone. **A retriever built on it returns arbitrary articles while producing perfectly plausible metrics** — the failure mode this probe exists to catch. MiniLM achieves a 348× larger margin from *half* the dimensions. Reported as a measured failure rather than a silent substitution.

4.4 Slate metrics barely discriminate — always show the random row [F25]

EB-NeRD slates average 11–12 items with exactly one click, so **random ranking scores nDCG@10 = 0.4367 and AUC = 0.4933**. The best retriever reaches 0.4750 / 0.5407. Every AUC sits within noise of 0.5. Any slate-regime table without a random baseline row is misleading, and recall@K in the corpus regime is the only metric that clearly separates retrievers here.

4.5 Conditional pooling earns its complexity on one dataset only

Pooling strategy counts over 4,000 impressions: EB-NeRD mean 3,991 / recent_half 9; MIND mean 2,117 / recent_half 1,462 / max_pool 270. **46% of MIND users have interests too scattered for a centroid, against 0.2% on EB-NeRD** — consistent with MIND’s shorter, more varied histories (median 19 vs 92 clicks).

4.6 One metric legitimately has no confidence interval

Q4.4 demands a bootstrap CI on every metric. **Coverage cannot honestly have one.** It counts *distinct* articles, so it grows with sample size; resampling n units with replacement makes ~37% duplicates, so every resample sees fewer unique articles. The first attempt produced 0.9783 [0.9035, 0.9235] — **the point estimate outside its own interval**. Subsampling without replacement fails at every ratio from 0.5 to 0.99. Reported as a point estimate marked (no CI): saying so beats shipping a plausible-looking number that is biased by construction.

5 Anti-gaming (Q9)

Leakage. History is truncated to clicks strictly before the impression time. `test_no_leakage.py` asserts this on the built store *and injects a deliberate post-boundary click to confirm the checker fails* — a test that passes both before and after the boundary breaks proves nothing. *Honest limit:* verifiable only where per-click timestamps exist, i.e. EB-NeRD. MIND’s history is untimestamped, so there the boundary rests on the authors’ construction; the store records `history_boundary_verifiable: false` rather than implying otherwise.

Serving-unavailable features. EB-NeRD’s training rows carry `read_time`, `scroll_percentage` and `next_read_time/next...` — all recorded *after* the click. Predicting a click from its own consequences is circular. The organisers settled the boundary: those four columns are exactly what is deleted from the test set. They are excluded in `clean.py` with a comment naming why. Our retrievers never touch them, so the honest with/without pair is the cold-start fallback flag, reported as a `features` column on every result row.

6 Where it breaks at 10×

- **The join breaks before the model does [F15].** EB-NeRD large is 52× small’s impressions and its history parquet exceeds its behaviours file. The data layer gives first.
- **The prediction path needed a different algorithm, not a bigger machine [F32].** Full-corpus retrieval per impression ran at 162 impressions/s (~4h/file) because it discarded 99% of each result. Scoring the slate directly: **2,570/s, 16×**.
- **Parallelism was memory-bound, and the fix was representation [F36].** A worker peaked at 15.1 GB, of which the index was only 1.8 GB — the rest was 116.8M click ids as Python `str`. Storing them as `array('i')` indices: **9.25 GB, and 45% faster**.
- **Swap does not slow the merge, it stops it [F38].** All 51 shards wrote at 6,426 lines/s, then the merge produced *zero* output for 20+ minutes with 24 GB swapped. The same merge with RAM free: **7 seconds**. Scoring streams and degrades gracefully; the merge hits a 13.3M-element set at random, where swap does not degrade — it stalls.
- **ANN is not yet needed.** Both corpora (<65K) are under the 200K threshold where brute force stops being affordable at 190–890 ms/query. At 10×, HNSW becomes necessary and its recall-vs-exact gap becomes the thing to report.

7 Leaderboards (Q5) and limitations

MIND submitted: **AUC 0.5568, MRR 0.2646, nDCG@5 0.2778, nDCG@10 0.3331**. EB-NeRD file generated and validated (13,336,711 lines, 100% valid permutations). *Screenshots overleaf*.

Limitations, stated plainly. (i) The offline harness was under-powered: at $n=800$ it reported MIND AUC 0.4981 [0.4776, 0.5190] while the leaderboard scored **0.5568** — *outside* the interval [F34]. Now $n=4,000+$, but several reported differences still have overlapping CIs and are labelled as such. (ii) Abstracts are 75% of the lexical index and buy **+0.011 recall against a ± 0.030 CI** — kept because Q2.1 mandates it, reported as the null result it is [F28]. (iii) HNSW figures come from *random* vectors (near-orthogonal, worst case) and are a pessimistic bound. (iv) MIND’s test window is a single day, so one day’s topical anomaly moves every MIND number.